

MARILYN CHAHINE

Machine Learning Research Engineer | Embodied AI • Multimodal AI

Paris, France | +33 7 82 99 36 40 | marilynychahine@hotmail.com | [LinkedIn](#) | [GitHub](#)

WORK EXPERIENCE

Research Intern – Vision-Language Models for Engagement Detection

May 2026 – Present

Institute of Intelligent Systems and Robotics (ISIR)

- Developing a vision-language model for interaction readiness prediction using SigLIP-based visual representations that fuse person-level observations with scene context.
- Evaluating the proposed approach against handcrafted cue-based baselines, to assess robustness to noisy observations.
- Designed and executed the complete annotation pipeline across ~1M frames from five public datasets, producing frame-level bounding boxes and interaction-readiness labels.

Research Intern – Applied ML for Perceptual Cognition

Jan 2025 – Jun 2025

Laboratory of Cognitive and Computational Neuroscience (LNC2)

- Built a large-scale behavioral data acquisition pipeline (400+ participants) enabling representation learning on human similarity judgments.
- Investigated and extended the VICE variational representation learning framework through model interpretation, dataset scaling analyses, and latent-space evaluation.
- Produced evidence that narrative information restructures learned latent embeddings from visual to semantic organization, extending representation learning to complex naturalistic scenes.

Research Intern – Multisensory Integration

Mar 2024 – Jun 2024

Institute of Intelligent Systems and Robotics (ISIR)

- Designed and conducted controlled psychophysical experiments combining custom 3D-printed haptic stimuli with visual textures to study multisensory perception.
- Developed Python analysis pipelines using psychometric modeling and statistical hypothesis testing to characterize visuo-haptic integration.
- Identified a visuo-haptic incongruence detection threshold ($\Delta_{\text{Hurst}} \approx 0.3\text{--}0.4$) and asymmetric perceptual biases, resulting in an IEEE World Haptics Conference Work-In-Progress Paper.

PROJECTS

- EEG World Models for Anomaly Detection:** Built JEPA-based models for EEG anomaly detection on TUH datasets (up to 89% AUROC) and explored anomaly detection from prediction energy using models trained solely on normal EEG signals (PyTorch).
- Social & Physical Affordances as Model-Based RL:** Reproduced and extended a model-based RL architecture integrating affordance perception with action selection (Python).
- Pose World Model:** Transformer-based world model for future human motion prediction from pose sequences, enabling anticipation and planning in embodied robotic systems (PyTorch).

PUBLICATIONS

- Chahine, M.,** Bailly, G., Gueorguiev, D.
Perceived Incongruence of Visual and Haptic Roughness is Asymmetric. IEEE World Haptics 2025 Conference.

EDUCATION

MSc. Machine Learning, Sorbonne Université

2025 – 2026

MSc. Cognitive Science, École Normale Supérieure – Université PSL

2023 – 2025

B. Eng. Computer Engineering, American University of Beirut

2019 – 2023

SKILLS

Programming Languages: Python, C++, Java, MATLAB

Frameworks & Tools: Git, PyTorch, ROS2, SQL, TensorFlow, W&B

Machine Learning: Computer Vision, Foundation Models, Generative AI, Multimodal Learning, Reinforcement Learning, Representation Learning, Robot Learning, Self-Supervised Learning

Other: Dataset Curation, Experimental Design, LaTeX

Languages: French (Fluent), English (Fluent), Arabic (Professional proficiency)